

PREDICTING STUDENT ANXIETY, STRESS AND DEPRESSION LEVELS

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1. Introduction

The purpose of this project is to use a mental health survey to predict the anxiety, stress, and depression levels of university students. It is not intended to provide a medical diagnosis. Rather, the models are used as an early screening tool that could help identify students who may benefit from professional support.

The GAD-7 for anxiety, PSS-10 for perceived stress, and PHQ-9 for depression are the three common screening measures used in the dataset. Each scale has a number of questions, a category label, and a total score.

The main goal is to determine whether the student's background and patterns from the other two scales can be used to predict the third mental health level.

2. Dataset Description

Property	Value
Data Points	1,977 students
Original column	37
Demographic fields	5 categorical fields
Question fields	26 numeric response
Total score fields	3 numeric totals
Output	3 categorical labels
Problem type	Multi class classification

	Age	Gender	Academic_Year	CGPA	Scholarship	GAD1	GAD2	GAD3	GAD4	GAD5	...	PHQ2	PHQ3	PHQ4	PHQ5	PHQ6	PHQ7	PHQ8	PHQ9	Depression_Value	Depression_Label
0	18-22	Female	Fourth Year or Equivalent	2.50 - 2.99	No	1	1	1	2	2	...	2	1	1	2	1	1	1	1	11	Moderate Depression
1	18-22	Male	First Year or Equivalent	3.80 - 4.00	No	2	2	1	1	1	...	1	1	1	1	1	1	1	1	9	Mild Depression
2	18-22	Male	First Year or Equivalent	3.00 - 3.39	No	2	1	1	0	2	...	0	2	3	2	2	2	2	1	16	Moderately Severe Depression
3	18-22	Male	First Year or Equivalent	3.40 - 3.79	No	2	1	1	1	1	...	1	1	1	1	1	1	1	1	9	Mild Depression
4	18-22	Male	First Year or Equivalent	3.40 - 3.79	No	1	1	1	1	1	...	1	1	1	1	1	1	1	1	9	Mild Depression

This is a classification problem because each output is a named level, such as Mild Anxiety or Severe Depression. The target is not a continuous number. Linear Regression was therefore not suitable as it predicts continuous values rather than class labels.

The five demographic fields are Age, Gender, Academic Year, CGPA and Scholarship. The remaining inputs are numeric questionnaire responses and total scores. The three targets contain 4 anxiety classes, 3 stress classes and 6 depression classes.

Why encoding is needed

Machine-learning models cannot directly perform calculations on text. For this reason, categorical values were converted into numbers. Ordered categories were mapped in their real order, while categories without a safe order were one-hot encoded.

2.1 How the Labels Are Built

For all 1,977 rows, Anxiety_Value equals the sum of GAD1-GAD7, Stress_Value equals the sum of PSS1-PSS10, and Depression_Value equals the sum of PHQ1-PHQ9. Each label is then assigned from score ranges.

Therefore, when a label is predicted, that scale's own questions and total are removed. Keeping them would give the model the answer. All three label columns are also excluded from every feature set.

```
Is each Value column simply the sum of its questions?
Anxiety_Value      matches in 100.0% of rows
Stress_Value       matches in 100.0% of rows
Depression_Value   matches in 100.0% of rows
```

Score range for each Depression level:

	min	max
Depression_Label		
Mild Depression	5	9
Minimal Depression	1	4
Moderate Depression	10	14
Moderately Severe Depression	15	19
No Depression	0	0
Severe Depression	20	27

So when a label is the target, its own questions and its own Value are dropped. The other two scales are kept, because predicting depression from anxiety and stress answers is a genuine prediction.

2.2 Correlation Analysis

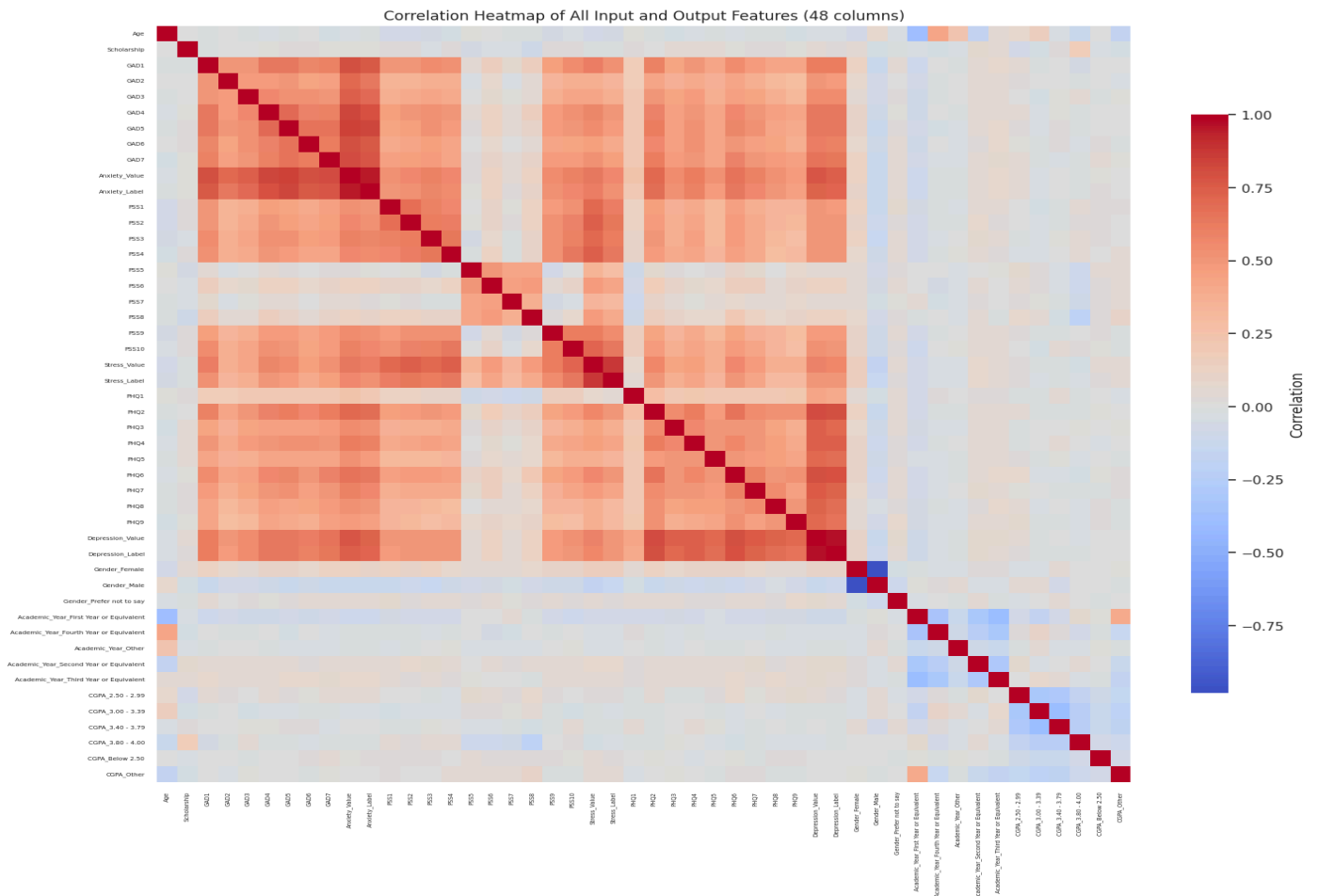


Figure 1: Correlation heatmap of all encoded input and output features.

The strongest relationships appear among the questionnaire scores, totals and their related labels. The depression total is strongly related to the anxiety total (about 0.768) and moderately related to the stress total (about 0.583). In contrast, most demographic correlations are close to zero. This suggests that the mental health scales carry useful predictive information, while background information alone is much less informative.

2.3 Imbalanced Dataset Check

The classes are not equally represented. This matters because a model can obtain a seemingly good accuracy simply by repeatedly choosing the largest class. The majority class baseline shows the accuracy achieved by that simple strategy.

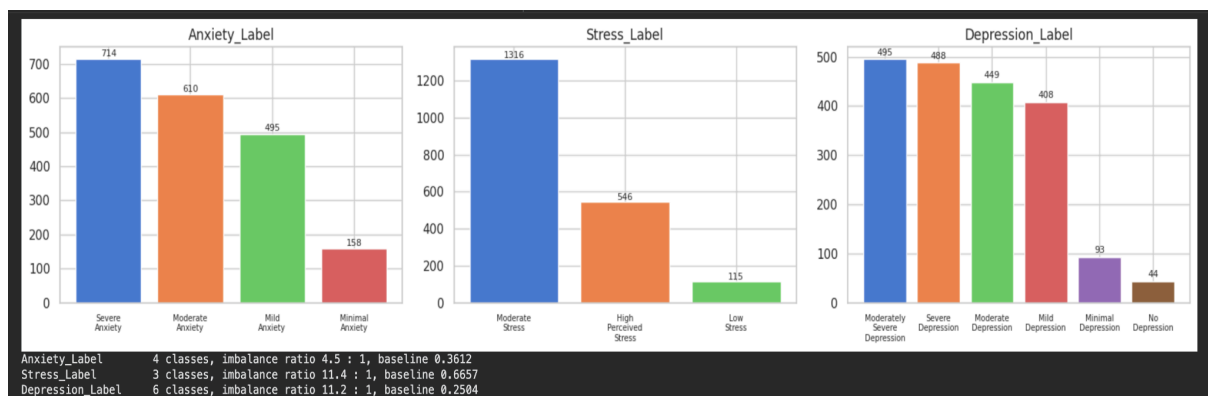


Figure 2: Class distribution for the anxiety, stress and depression targets.

Target	Classes	Class counts (Largest to smallest)	Ratio	Baseline
Anxiety	4	714 / 610 / 495 / 158	4.5 : 1	0.3612
Stress	3	1316 / 546 / 115	11.4 : 1	0.6657
Depression	6	495 / 488 / 449 / 408 / 93 / 44	11.2 : 1	0.2504

Stress is the most visibly imbalanced target. Moderate Stress contains about two thirds of the students. For that reason, accuracy is reported together with weighted precision, weighted recall, confusion matrices and AUC. A result should be compared with its own target baseline, not with the baseline of another target.

2.4 Exploratory Data Analysis

Exploratory data analysis was used to check relationships before training the models. The top two boxplots compare anxiety and stress scores with the six depression levels. The bottom charts compare depression levels by gender and academic year.

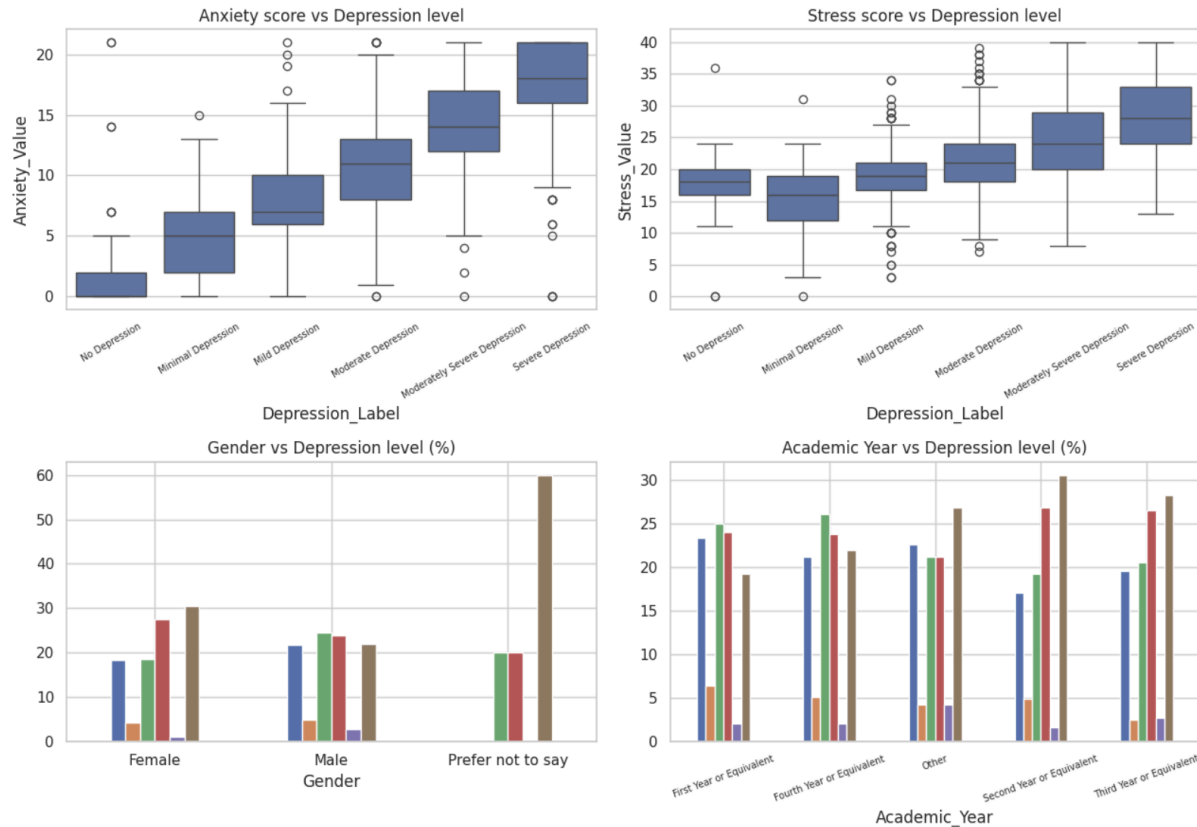


Figure 3: Important relationships found during exploratory data analysis.

- Anxiety scores generally rise as the depression level becomes more severe.
- Stress scores also rise across the depression levels, although the pattern is less steep.
- Gender and academic year show much flatter differences, so they are weaker predictors on their own.

The EDA supports the correlation result that related mental-health symptoms provide the clearest signal in this dataset.

3. Dataset Pre-processing

3.1 Problem 1: Missing Values

Only six values were missing, all in the Gender column. Deleting those students was unnecessary because the missing part was very small.

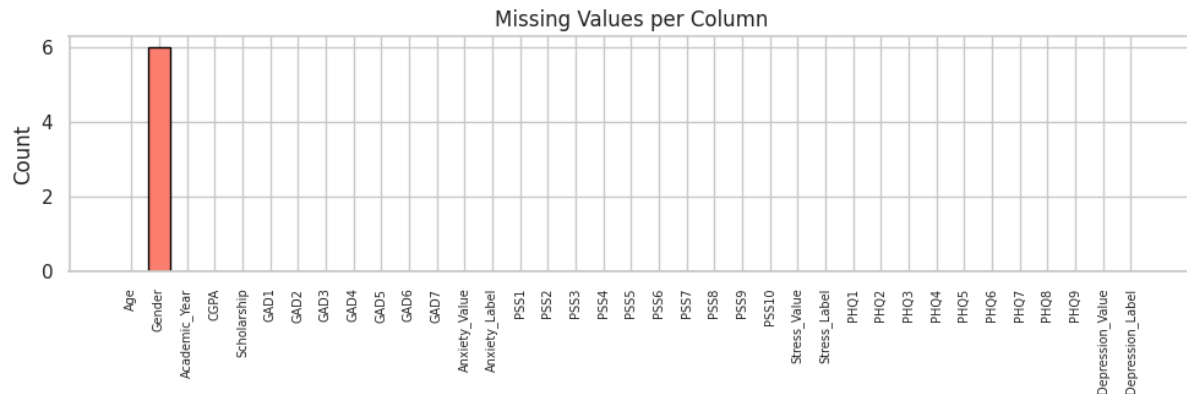


Figure 4: Missing values before imputation.

The six values were replaced with the mode, which was Male, the most frequent category (1,369 records). This kept all 1,977 students and left no missing values.

3.2 Problem 2: Categorical Values

Several columns contained text that the models could not calculate with directly. Age has a real order, so the groups were mapped from youngest to oldest. Scholarship has two values and was changed to No = 0 and Yes = 1.

Gender was one-hot encoded because it has no order. Academic_Year and CGPA were also one-hot encoded because both include an Other category, which has no safe numerical rank. The target labels were mapped from the lowest to the highest severity. After one-hot encoding, the 37 original columns became 48 numeric columns.

3.3 Problem 3: Feature Scaling

The inputs use different ranges: individual questions are mostly 0-3 or 0-4, while the totals can reach 21, 40 and 27. Without scaling, the larger ranges would have too much influence on distance and gradient calculations.

StandardScaler changed each training feature to mean 0 and standard deviation 1. It was fitted on the training set only, then applied to both training and test data. This avoids allowing test information to influence training.

4. Dataset Splitting

Each target used a stratified 80% training and 20% test split: 1,581 records for training and 396 for testing. Stratification kept the class proportions similar in both parts. `random_state = 42` fixed the split so the same experiment could be repeated.

Anxiety_Label	features=37	train=1581	test=396	baseline=0.3611
Stress_Label	features=34	train=1581	test=396	baseline=0.6667
Depression_Label	features=35	train=1581	test=396	baseline=0.2500

The final input sizes were 37 features for Anxiety, 34 for Stress and 35 for Depression. They differ because each target's own questions and total were removed.

5. Model Training and Testing

The project used KNN, Logistic Regression and a Neural Network. K-Means was added for the required unsupervised analysis.

The template also lists Decision Tree, Naive Bayes and Linear Regression as alternatives. Linear Regression does not fit categorical targets, and the strongly related questionnaire features make the Naive Bayes independence assumption less suitable. KNN and Logistic Regression already satisfy the two-model requirement with clearly different methods.

5.1 K-Nearest Neighbours (KNN)

KNN assigns a student the most common label among the nearest training records. Odd values from $k = 1$ to 31 were checked, and $k = 15$ was used because test performance became stable after about $k = 10$.

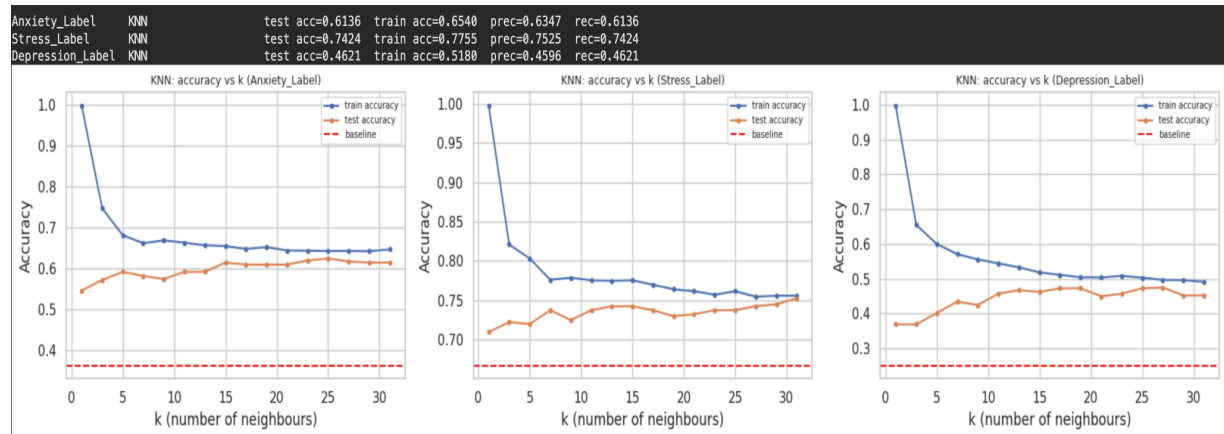


Figure 5: KNN training and test accuracy across different values of k .

KNN reached 0.6136 accuracy for Anxiety, 0.7424 for Stress and 0.4621 for Depression. All three results were above their baselines. Training accuracy was higher than test accuracy, especially for small k , because a small neighbourhood memorises individual records more easily.

5.2 Logistic Regression

Logistic Regression estimates a probability for each class using a linear combination of the input features. Its train and test results were close, so it generalised more consistently than the neural network.

Anxiety: test accuracy 0.6136; training accuracy 0.6648.

Stress: test accuracy 0.7879; training accuracy 0.7685.

Depression: test accuracy 0.5303; training accuracy 0.5174.

Anxiety_Label	Logistic Regression	test acc=0.6136	train acc=0.6648	prec=0.6074	rec=0.6136
Stress_Label	Logistic Regression	test acc=0.7879	train acc=0.7685	prec=0.7764	rec=0.7879
Depression_Label	Logistic Regression	test acc=0.5303	train acc=0.5174	prec=0.5205	rec=0.5303

Train and test scores are close together, so this model does not overfit.

5.3 Neural Network

The network used 64 ReLU units, Dropout(0.3), 32 ReLU units and a softmax output. It was trained with Adam and sparse categorical cross-entropy for 40 epochs, a batch size of 32 and a 15% validation split.

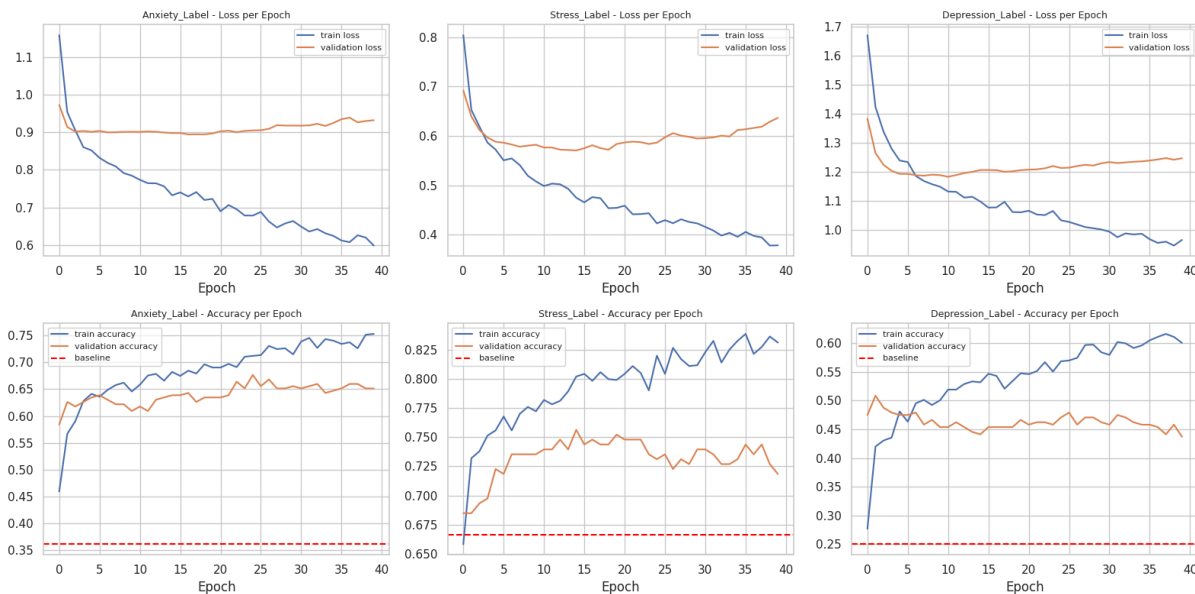


Figure 6: Neural-network training and validation loss and accuracy across 40 epochs.

The network reached test accuracies of 0.614 for Anxiety, 0.753 for Stress and 0.470 for Depression. In every case the training curve continues improving more than the validation curve. This gap shows overfitting as the network starts to learn details specific to the training records. Dropout reduces the problem but does not remove it completely.

5.4 K-Means Clustering (Unsupervised)

K-Means ignores the known labels and divides students into groups based only on similarity. The depression feature set was used, and $k = 6$ was selected so the number of clusters matched the six depression levels. PCA reduced the features to two dimensions only for visualisation.

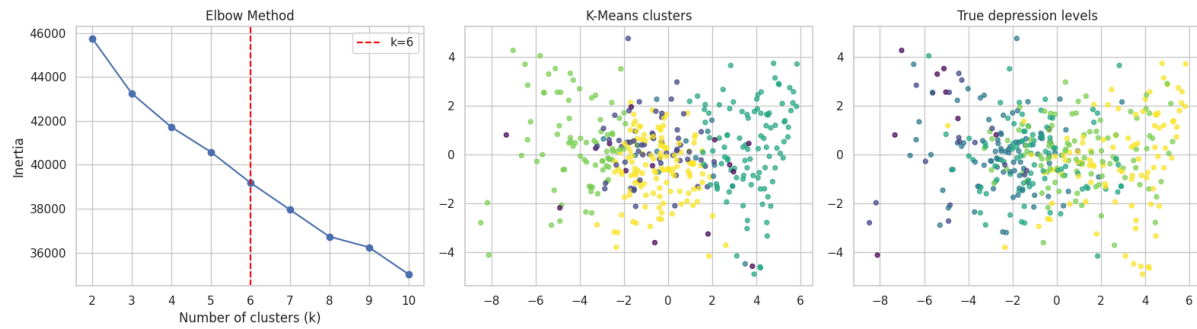


Figure 7: Elbow curve, K-Means clusters and true depression levels after PCA projection.

The Adjusted Rand Index (ARI) was 0.1296. An ARI of 1 means perfect agreement and a value near 0 means chance-level agreement. Therefore, the clusters share only a weak relationship with the real depression levels. This is reasonable because K-Means groups overall patterns, while the target labels are created from specific depression score ranges.

The elbow graph also does not show one sharp turning point at $k = 6$. The choice of six clusters was made for direct comparison with the six known depression categories, not because the data naturally formed exactly six clear groups.

6. Model Selection and Comparison

6.1 Accuracy, Precision and Recall

Target	Model	Accuracy	Precision	Recall	Baseline
Anxiety	KNN	0.6136	0.6347	0.6136	0.3611
Anxiety	Logistic Regression	0.6136	0.6074	0.6136	0.3611
Anxiety	Neural Network	0.6136	0.6135	0.6136	0.3611
Stress	KNN	0.7424	0.7525	0.7424	0.6667
Stress	Logistic Regression	0.7879	0.7764	0.7879	0.6667
Stress	Neural Network	0.7525	0.7417	0.7525	0.6667
Depression	KNN	0.4621	0.4596	0.4621	0.2500
Depression	Logistic Regression	0.5303	0.5205	0.5303	0.2500
Depression	Neural Network	0.4697	0.4578	0.4697	0.2500

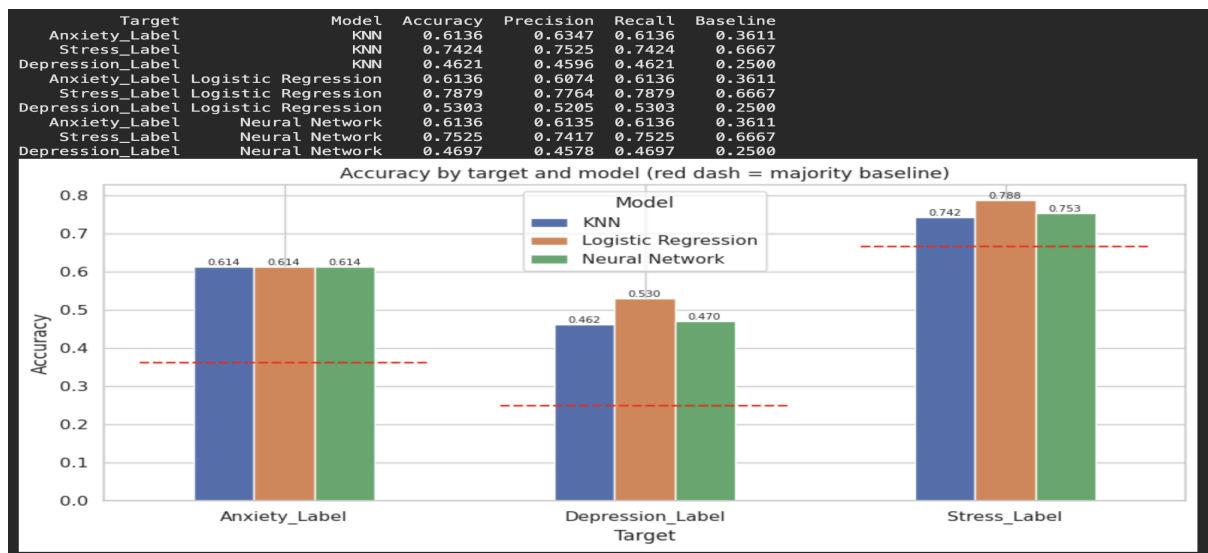


Figure 8: Accuracy comparison. Each red dashed line is that target's majority baseline.

All three models tied at 0.614 for Anxiety. Logistic Regression was best for Stress at 0.788 and for Depression at 0.530. Every result exceeded its own majority baseline, although the improvement for Stress was smaller because its baseline was already high at 0.667.

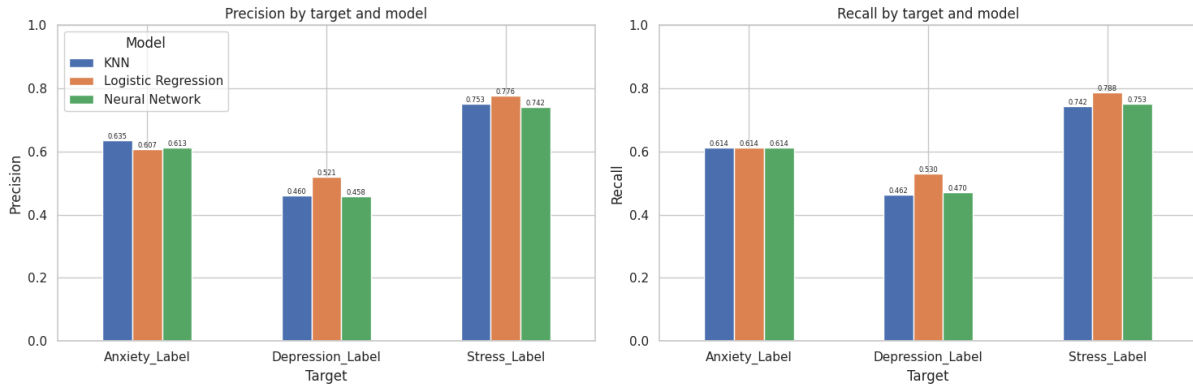


Figure 9: Weighted precision and recall for every target and model.

Precision and recall follow the same general pattern. Logistic Regression leads for Stress and Depression, while KNN has the highest Anxiety precision. Weighted recall equals accuracy here because every record has exactly one true class.

6.2 Confusion Matrices

Rows are the true levels and columns are the predicted levels. Correct predictions appear on the main diagonal. The other cells show where one level was confused with another.

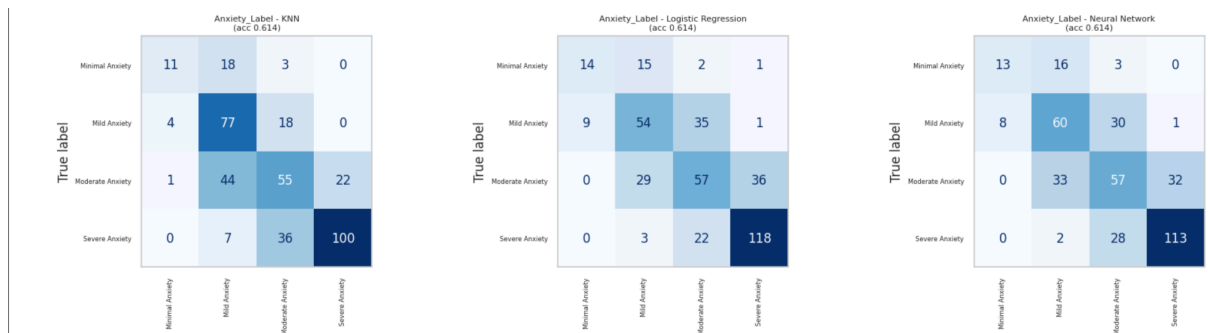


Figure 10: Confusion matrices for the Anxiety target.

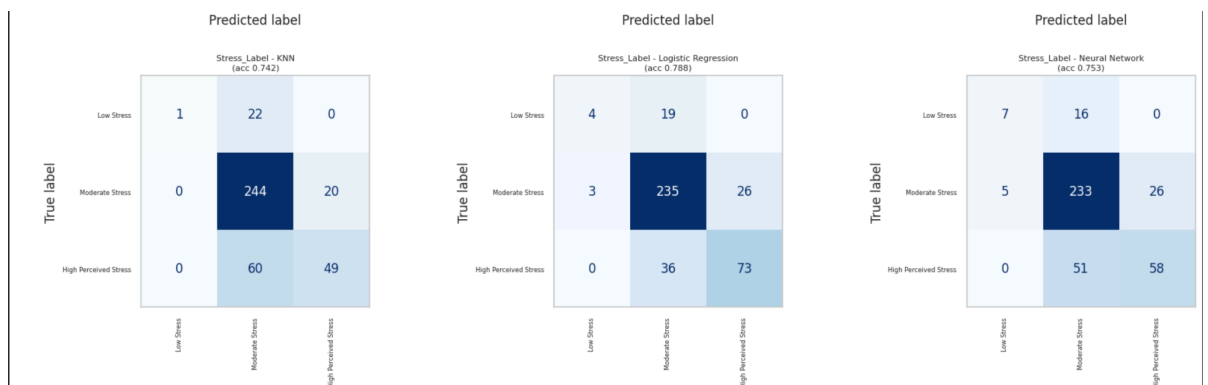


Figure 11: Confusion matrices for the Stress target.

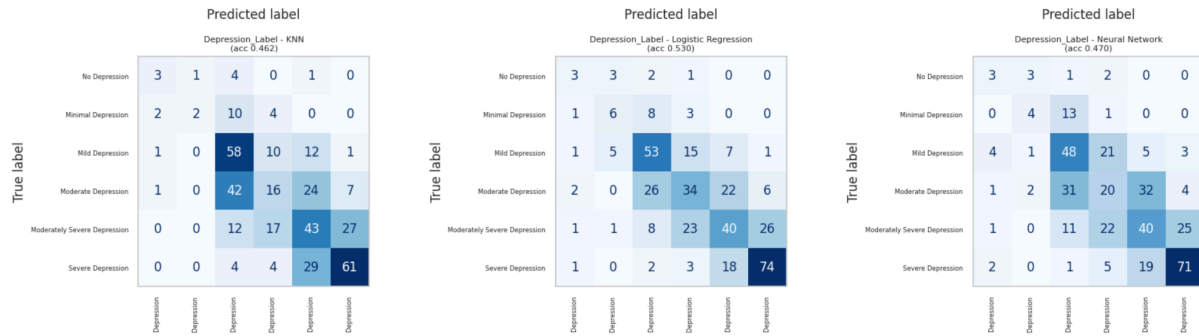


Figure 12: Confusion matrices for the Depression target.

Most errors occur between nearby severity levels. This makes sense because the categories are created by cutting continuous questionnaire totals into ranges. The smallest groups, especially No Depression and Minimal Depression, remain the hardest to predict.

6.3 ROC Curves and AUC

For the multi-class targets, one-vs-rest ROC curves were calculated and then macro-averaged. AUC = 0.5 represents random ranking, while a higher value means better separation between the levels.

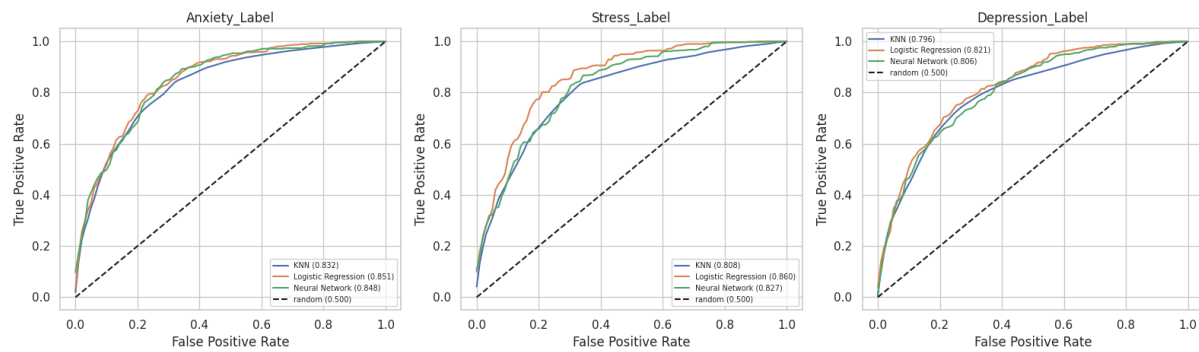


Figure 13: Macro-averaged one-vs-rest ROC curves for all targets and models.

Logistic Regression has the highest AUC for all three targets. Every AUC is well above 0.500, confirming that the models learned useful ranking information even when exact class prediction remained difficult.

7. Conclusion

Results

The project successfully trained three supervised models for Anxiety, Stress and Depression and also completed an unsupervised K-Means analysis. Every supervised result beat its own majority-class baseline. For Anxiety, all three models tied at 0.614 compared with a baseline of 0.361. Logistic Regression performed best for Stress with 0.788 accuracy against 0.667, and for Depression with 0.530 accuracy against 0.250.

Logistic Regression was the most reliable overall model. It produced the best AUC for all three targets (0.851, 0.860 and 0.821) and the best exact-class accuracy for Stress and Depression. KNN also performed above baseline and tied for the best Anxiety accuracy. The neural network learned useful patterns, but its larger training-test gaps show that it overfitted more than the other models.

Reasoning for the Results

1. The correlation analysis found strong relationships among the three mental-health scales, especially between anxiety and depression.
2. The EDA boxplots showed anxiety and stress scores increasing with depression severity.
3. Every AUC value was between 0.796 and 0.860, clearly above random guessing at 0.500.
4. K-Means achieved only 0.1296 ARI, showing that natural clusters overlap with depression levels only weakly.

Together, these results suggest that related symptom scores are useful for predicting another mental-health level, while demographic information alone contributes much less. The results should still be treated as screening evidence rather than a clinical diagnosis.

Comments on Model Performance

- Logistic Regression generalised most consistently because its train and test results stayed close.
- KNN showed some memorisation, particularly for small values of k , but became more stable around $k = 15$.
- The neural network's training performance continued improving while validation performance flattened, which is a clear sign of overfitting.
- Class imbalance made the smallest classes difficult to predict, so accuracy was not used alone.

Challenges

- Handling the 6 missing Gender values without removing useful student records.
- Selecting different encodings for ordered, binary and unordered categorical variables.
- Preventing leakage by removing each target's own questions and total, removing all label columns, and fitting the scaler only on training data.
- Interpreting imbalanced multi-class results using baselines, weighted metrics, confusion matrices and AUC.